

A Déjà-View of the Interpretability Illusion in BERT

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Abstract

In this work, we document the reproduction steps of Bolukbasi et al. (2021), confirming the initial findings that: 1. an interpretability illusion exists in BERT, and 2. that this illusion could be explained by a combination of causes, including dataset idiosyncrasy, geometric patterns in the embedding space, and human error. Additionally, our findings show that using an LLM as a judge for the presence/absence of patterns in top-activating sentences yields results consistent with humans for items expected to be generally meaningful, while completely rejecting highlighting patterns in random noise, in which overeager humans are more inclined to find nonexistent common threads.

1 Introduction

Studies in computer vision suggest that neurons in neural networks trained for image recognition encode meaningful patterns, such as lines, shapes, or geometric motifs (Bau et al., 2017; Hernandez et al., 2022). A natural hypothesis to test is whether neurons in bidirectional text encoders can be interpreted as having human-recognizable patterns to their activation.

The study of Bolukbasi et al. (2021) suggests that this pattern found in image recognition models does not generalize to bidirectional encoders, and we set our task to reproduce their results. The central idea is to use BERT over a large sample of sentences from multiple datasets and to analyze the sentences that most activated certain target neurons in the last layer. The activations of these neurons can be indicators of patterns related to interpretability. If top-activating sentences of that neuron have an identifiable pattern, then it can be said that the neuron has “learned” that pattern. However, when changing the dataset, we observe that target neurons are activated by sentences exhibiting other apparent patterns or, more concerningly, no pattern

at all. We look for concrete evidence that the apparent interpretability of these neurons is, in fact, illusory, and to investigate some possible causes of this illusion. All material used and produced will be made publicly available under CC-BY 4.0 license.¹

2 Datasets

We use 4 datasets in our experiments, covering approximately 600k sentences. These datasets are as follows: 1. **QQP**: Questions posted on Quora; 2. **QNLI**: A Wikipedia-based dataset with sentences from the Question-answering Natural Language Inference dataset; 3. **Wiki**: A Wikipedia-based dataset with sentences directly collected from articles; 4. **Books**: Sentences extracted from diverse fiction books. The samples used in our experiments are from the training splits. Pseudo-random number generation for dataset sampling, as well as for all pseudo-random operations in our code, used the fixed seed 1.

2.1 QQP

For Quora Questions, we use the same dataset as the one used by the original paper’s authors, namely AlekseyKorshuk/quora-question-pairs.² This dataset was initially created for text classification, in which the aim was to train a model to determine whether two sentences are equivalent in meaning (Galke et al., 2022). The sentences are grouped in pairs and one sentence may be found in multiple pairs. We perform de-duplication based on the sentence IDs provided in the dataset, then sample 164.246 sentences from among them. Example sentences from this dataset include:

- *How do I log out of my Gmail account on my friend’s phone?*

¹<https://anonymous.4open.science/r/blackbox-re26-anon/>

²<https://huggingface.co/datasets/AlekseyKorshuk/quora-question-pairs>

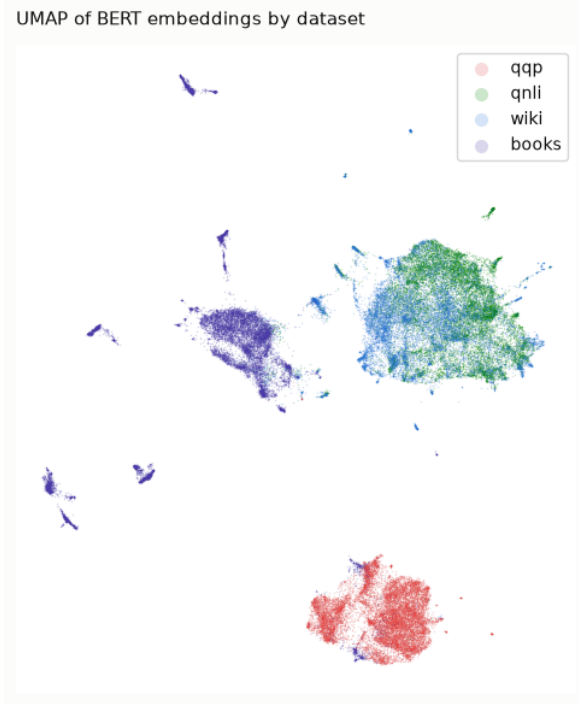


Figure 1: Visualization of datasets in the embedding space after UMAP. Note the closeness of the Wikipedia-based datasets, while Books and QQP are relatively separate.

- *When do you use ㄝ instead of ㄣ?*

2.2 QNLI

We sourced this dataset from SetFit/qnli.³ It comprises both questions and answers, as well as some other extraneous fields. We use only the *answers* field, remove null values, and extract a set of 58.644 sentences. This is off-by-one from the initial paper’s 58.645 sentences, which is explained by the `dropna()` call’s removal of exactly one null value. Example sentences in this dataset include:

- *Around the same time, another of Sulla’s former lieutenants, Marcus Licinius Crassus, had just put down the Spartacus-led gladiator/slave revolt in Italy.*
- *Research since reveals self-examination beginning early in adolescence, but identity achievement rarely occurring before age 18.*

2.3 Wiki

We sourced this dataset from google/wiki40b.⁴ For this dataset, Bolukbasi et al. mention that preparation was done in the same manner as the

³<https://huggingface.co/datasets/SetFit/qnli>

⁴<https://huggingface.co/datasets/google/wiki40b>

BERT paper of Devlin et al. (2018). Because no official dataset was available, we selected this dataset after observing that it does not contain lists, tables, or section headers. Article texts are delimited with separators introduced by the maintainers in the text.

Another matter with this dataset is extracting sentences from contiguous blocks of prose, which is the form in which Wikipedia articles are presented. To this end, we used spaCy (Honnibal et al., 2020) with the English `en_core_web_sm` model to sentence-split the articles and extract 203.736 sentences. This model has been used for similar tasks previously (Zanotto and Aroyehun, 2025; Hossain et al., 2026). Example sentences from this dataset include:

- *Instead, Erasmus compared God to an astronomer who knows that a solar eclipse is going to occur.*
- *After a crowdfunding campaign, the cafe, at Old Christchurch Road, will have 12 resident rescue cats roaming the two floors.*

2.4 Books

For this dataset, we sourced from the repository SamuelYang/bookcorpus⁵ and sampled 198.085 sentences from it without any further processing. This dataset has been previously used in a paper on autoencoders (Liu and Yang, 2022) and contains approximately 74 million sentences extracted from works of fiction. Example sentences from this dataset include:

- *It takes more time to draw a sword than it does to get struck.*
- *I can run them through a program that flags image manipulation.*

3 Embeddings

We use the base uncased BERT model⁶ to obtain sentence embeddings. BERT’s tokenization of a sentence includes two features that are not directly sourced from the original text: 1. a postpended sentence termination token, and 2. a prepended [CLS] token representing the sentence as a whole. The activation of last-layer neurons on this prepended token is taken as the embedding of a sentence.

Generating embeddings is a computationally expensive task. We have found that a naive call to the tokenizer followed by a call to BERT produces

⁵<https://huggingface.co/datasets/SamuelYang/bookcorpus>

⁶<https://huggingface.co/google-bert/bert-base-uncased>

around 2-3 sentence embeddings per second, resulting in an impractical embedding time on an average Google Colab. By enabling `truncation=True` in the tokenizer call and disabling the part of BERT that calculates gradients for backpropagation, we were able to produce the embeddings in around 20 minutes on a consumer-grade GPU GeForce RTX 3050 Ti.

3.1 Use of NNsight

We integrated NNsight into our workflow and confirmed that sentence embeddings produced with it are qualitatively identical to those produced by simply extracting values from the hidden layers variable of BERT. While its usefulness in generalizing to other models cannot be overstated, we must note the naming convention it introduces. Specifically, the output field of the `LanguageModel` object representing BERT is renamed to `nns_output`, since BERT already uses the name `output`. Using NNsight is only about 4 times slower than extracting sentence embeddings directly, which is acceptable for what is a one-time step.⁷

3.2 Exact embedding matches

We note that the generated embeddings will never be identical, even if a fixed-seed pseudo-random number generator is used. The loss of precision is due to floating-point arithmetic, which is essentially performed arbitrarily by the processing unit. This representation of real numbers lacks properties that may otherwise be taken for granted; e.g., $a + (b + c)$ is not equal to $(a + b) + c$ in this arithmetic. We have found a maximum absolute difference on the order of 10^{-6} and a mean difference on the order of 10^{-8} between a sample of sentence embeddings generated in one pass with NNsight and the same sentence embeddings extracted by raw model access, which is consistent with floating-point precision loss.

4 The Interpretability Illusion

In order to show that the patterns found by humans among the top-activating sentences for the neurons of BERT are illusory, we must first establish that humans do actually find meaningful patterns among them. To this end, we extract panels of $k = 10$ sentences that are prepared further for human annotation. To obtain the panels, we take the top-10

activating sentences for 25 neurons and 33 random directions across each of the 4 datasets. Thus, we produced 29 sets of random sentences, 8 for QQP, and 7 for each other dataset, which were used for evaluation, resulting in a total of 2610 sentences.

4.1 Random Directions

The output space of BERT for one token is 768-dimensional, with one component indicating the activation of each neuron. To obtain a random direction, we draw 768 values from a `standard_normal` distribution. We apply normalization after all the directions are generated. The idea is that there may be some directions in this space along which more and more sentences that share a pattern appear. The score for a sentence s with embedding e_s along a direction d_i is the dot product $d_i \cdot e_s$, and the top k sentences for this direction are considered to be those having the highest k scores.

4.2 Sentence Annotations

The entire set of 2610 sentences is sent to manual human annotators. We instruct two annotators to write whichever patterns they perceive as being representative of a sentence. Annotators for this experiment are allowed to write anything they find to be a pattern shared by the sentences of the panel. We manually group annotations that are identical in meaning afterwards. A total of 261 panels were annotated, split between 2 human annotators with no overlap to produce 1 set of complete annotations. Each annotator took approximately 4 hours to complete their set. Annotators were blinded to source and condition and were shown panels successively, with the possibility to review and correct their previous annotations.

For example, we take a look at direction 28 in the QQP dataset to offer an example of an annotation task. The top-3 sentences⁸ in this panel are:

- *Which is the best 50" Samsung 3D LED TV?*
- *Samsung Galaxy Note 4: Will the official update for Android Mar 6.0 come to Note 4 and 3?*⁹
- *What are the best smartphones under 10000 in 2016?*

For this panel, our annotator produced 1 label,

⁸The entire panel, with the other 7 sentences, may be found in our repository.

⁹Spelling and punctuation were fixed when reproducing these particular sentences in this paper for readability. For annotation, the original sentences were shown every time.

⁷We are considering a pull request in the NNsight library to address these issues.

namely “technology.” Had multiple patterns, or no patterns, been found, the annotator would have been able to specify this fact.

4.3 Results

Annotators clearly identify patterns in the dataset, labeling on occasion even multiple patterns for one panel, as shown by Table 1.

Condition	Panel count	μ	σ
BERT neurons	100	1.73	1.01
Random directions	132	1.45	0.81
Random sentences	29	0.21	0.49

Table 1: We collected annotation results and counted how many patterns were found for each panel, separating panels based on whether they came from neurons, randomly-generated directions or were panels of random sentences. Then, we find the mean and variance for each of these 3 conditions. Values over 1 are expected, since annotators may find multiple patterns in one panel.

Patterns are found to be present across all three experimental conditions. While more prevalent in neurons and the randomly-generated directions in the embedding space (see Table 2), they are also found in sets of random sentences.

Condition	Panel count	Pattern %
BERT neurons	100	87.00
Random directions	132	88.64
Random sentences	29	17.24

Table 2: We review annotation data for the three types of panel (neuron, random direction, random sentences) and find the percentage among them for which patterns were identified. Neurons and random directions are significantly more pattern-annotated than random sentences.

5 Experiments

5.1 Dataset Idiosyncrasies

A first hypothesis is that datasets are separable in the embedding space. The UMAP diagram (McInnes et al., 2018) in Figure 1 confirms this finding. We sampled 40,000 sentences from each dataset and trained a linear SVM classifier with $C = 1$ on an evenly-stratified train-test split of 20% produced from these sentences. The classifier was trained with input data containing 768 features, one for the activation of each neuron on the [CLS] token of the sentence. That is to say, our classifier used the sentence embeddings directly as input.

Our classifier has little trouble separating QQP and Books, and the mistakes made with regard to Wiki and QNLI are, to some extent, forgivable, as the datasets are sourced from the same material (see Figure 2).

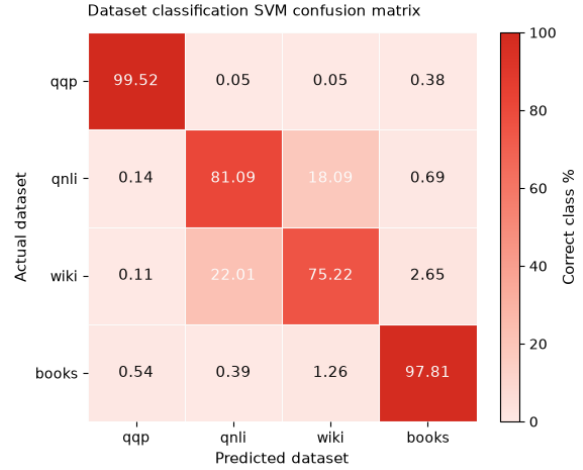


Figure 2: Confusion matrix of a SVM for source dataset identification.

5.2 Semantic Coherence

A second hypothesis is that the interpretability illusion could also arise due to the presence of concepts, which are topics, themes, or, more generally, semantic properties of the sentences in the dataset. These can be defined by three criteria. **Global** concepts and **dataset-level** concepts are measured based on the presence of certain words that represent them. If the frequency of a word increases along a direction in BERT’s embedding space, or along a direction in a dataset, then we have found evidence of the global or dataset-level concept, given that the token exists. **Local** concepts represent the idea that similar sentences are clustered together while not having a direction for the frequency of concept occurrence to increase along.

5.2.1 Identifying Monotonic Tokens

We start by defining a vocabulary of words to consider. A word is included in the vocabulary if it appears at least 100 times in each of the datasets.¹⁰ To prove the existence of these concepts, for each word in this vocabulary, we sort the sentences from least to most activating of each of our 25 annotated neurons, then measure how many of those sentences the word occurred in, grouping them into

¹⁰We identify 828 words with this property.

quintiles. If a word tends to be absent when the neuron is inactive and present when the neuron is active, then we may take it as evidence that the concept is encoded by the neuron.

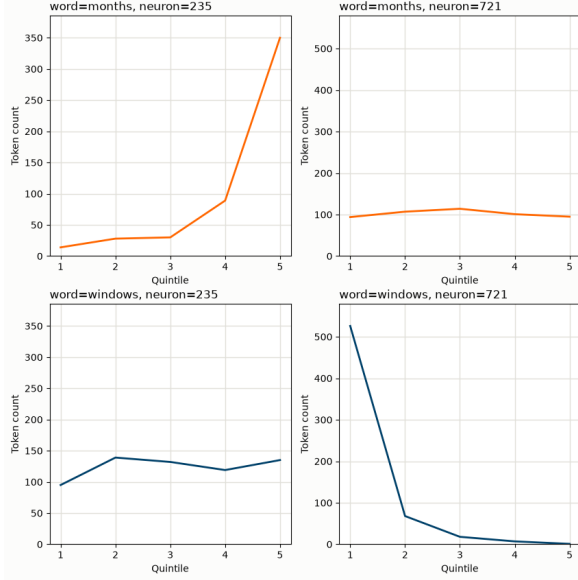


Figure 3: We measured the presence of tokens months and windows among the top-activating sentences of neurons 235 and 721. A word strongly associated with one neuron is found to be unremarkable when considering the activations of most other neurons.

Our results are numerically consistent with Bolukbasi et al. for this experiment. We note the presence of the same patterns in our values. For instance, Table 3 shows that the combination of datasets QNLI (labeled 2) and Wiki (labeled 3) has a higher rate of monotonic words than other combinations of two datasets. We attribute this to their closeness in the embedding space, recalling that they are both based on content from the same source. These values are consistent with the presence of global and dataset-level patterns and inconsistent with random chance. Assuming the null hypothesis, a reasonable model would be to say that between two quintiles, the probability of the occurrence count increasing or decreasing is 50%. Thus, we would observe, say, monotonically increasing values with a probability of 6.25% for one dataset. The difference is easier to highlight when considering combinations of datasets. Assuming independence, we would expect that around 0.00153% of words will be monotonically increasing by random chance across all four datasets, which is significantly lower than the observed 0.74%.

Datasets	Monotonic	Increasing	Decreasing
1	25.85%	12.86%	12.99%
2	19.56%	9.71%	9.85%
3	30.30%	15.01%	15.28%
4	25.17%	12.57%	12.60%
1, 2	6.53%	2.75%	2.77%
1, 3	9.55%	3.99%	4.06%
1, 4	8.14%	3.40%	3.41%
2, 3	10.75%	5.27%	5.36%
2, 4	6.11%	2.41%	2.43%
3, 4	9.43%	3.92%	3.96%
1, 2, 3	4.04%	1.77%	1.80%
1, 2, 4	2.48%	0.99%	1.00%
1, 3, 4	3.65%	1.50%	1.51%
2, 3, 4	3.85%	1.64%	1.66%
1, 2, 3, 4	1.72%	0.74%	0.75%

Table 3: Percentages of monotonic tokens across neuron activation quintiles, considering each combination of datasets. Datasets were numbered 1 for QQP, 2 for QNLI, 3 for Wiki and 4 for Books.

5.2.2 Local Concepts

We also test the hypothesis that the observed sentence patterns arise from local semantic structure rather than from a semantically meaningful direction. In this experiment, we refer to both arbitrary directions and neurons simply as *directions*, since both can be represented as vectors in the embedding space. Random directions are randomly sampled vectors, whereas individual neurons correspond to the standard basis vectors of the embedding space.

For each direction p , we select the $k = 10$ sentences with the highest activations. We denote this set by $S_{p,k}$. For each sentence $s \in S_{p,k}$, we find its k most similar sentences in the original embedding space:

$$N_k(s) = \arg \max_{\substack{S' \subseteq S \setminus \{s\} \\ |S'|=k}} \sum_{\hat{s} \in S'} e_s \cdot e_{\hat{s}},$$

where e_s is the embedding of sentence s , and S is the set of all sentences. Three similarity distributions are made: 1. $D_{p,\text{nearest}}$: similarities between each top-activating sentence and its nearest neighbors; 2. $D_{p,\text{top}}$: similarities between pairs of top-activating sentences; 3. $D_{p,\text{random}}$: similarities between top-activating sentences and randomly selected sentences. If direction p represents a meaningful concept, we expect the top-activating sentences to be similar to one another. There-

fore, $D_{p,\text{top}}$ should be more similar to $D_{p,\text{nearest}}$ than to $D_{p,\text{random}}$. We compare the distributions by converting them into histograms and using the weighted Jaccard similarity:

$$L(h_1, h_2) = \frac{\sum_i \min(h_1(i), h_2(i))}{\sum_i \max(h_1(i), h_2(i))}.$$

The locality score for direction p is

$$L_p = L(D_{p,\text{nearest}}, D_{p,\text{top}}).$$

The score ranges from 0 to 1. A high score (meaningful) means that the top-activating sentences have a similarity pattern close to that of local neighbors, while a low score (meaningless) means that the two distributions have little overlap.

Dataset	Meaningful	Meaningless	p
All	0.0433	0.0101	0.0004
QQP	0.0148	0.0040	0.0147
QNLI	0.0410	0.0113	0.4760
Wiki	0.0488	0.0189	0.3037
Books	0.0629	0.0226	0.4071

Table 4: Locality scores for neurons considered meaningful and meaningless across datasets. We attribute the high p -values of some datasets due to the small number of meaningless directions. QNLI has only 5 meaningless examples, and Wiki and Books have 4 each.

Table 4 contains the per-sentence locality scores across datasets with additional p -values reported for a Mann–Whitney U test. Although the differences are not statistically significant for some individual datasets, this may be due to the small number of directions classified as meaningless, which reduces the statistical power of the test. When all datasets are combined, the difference becomes statistically significant. The difference between the meaningful and meaningless direction locality scores observed is around 3-4 times, which is slightly larger than in the original paper. Indeed, if annotations are fairly liberal, then it would be reasonable to expect the few meaningless directions to be truly meaningless.

5.3 Annotation Issues

Given that the annotation task has open-ended answers, annotator errors can lead to a possible reason for why the interpretability illusion appears. The hypothesis that imaginative annotators find illusory patterns seems plausible at first glance. We attempt to offer a novel perspective supporting the idea that

the interpretability illusion arises from the documented human tendency to fill in patterns where there are none. As such, we use Claude Opus 5 with the effort setting on Max to do the sentence annotation.

Condition	Panel count	Pattern %
BERT neurons	100	76.00
Random directions	132	80.30
Random sentences	29	00.00

Table 5: Pattern occurrence rate across experimental conditions as resulting from data annotated by an LLM. We remark the score of 0% on random patterns, suggesting that human labelings of random sentences might sooner be explained by apophenia rather than global concepts.

This annotation results appears to be largely consistent with human-annotated data, both in Bolukbasi et al. and in our experiment, finding patterns in BERT neurons and random directions at a rate slightly less than 80%, which is consistent with a stricter annotator. The tendency to find slightly more patterns in random directions than in BERT neurons is also present here. However, these annotations differ in that the LLM finds no pattern at all in any of the random sets of sentences (see Table 5), suggesting that human error is a notable part of the interpretability illusion.

6 Conclusions

The main findings of our study indicate that the interpretability illusion in the uncased BERT model is a real and reproducible phenomenon. Furthermore, we explain this phenomenon with arguments based on dataset idiosyncrasies, the geometry of the embedding space, and human error in identifying patterns and concepts in arbitrary sentences.

Future work on this subject would involve exploring the interpretability illusion for other models and larger collection of datasets. Furthermore, we may continue exploring the idea of using an LLM as judge, possibly leading to comparative work on what patterns humans and LLMs agree and disagree upon in panels of sentences. It may also prove interesting to do a comprehensive analysis of neurons on intermediary layers to see if similar numerical results are obtained. If that were the case, it would serve as further evidence that any apparent interpretability is in fact illusory.

7 Limitations

There are a few limitations to be acknowledged. The lack of overlap between the panels that the annotators were given makes it impossible to score conflict between said annotators, conflict which was recorded in a table in the original paper. Despite this, we believe that our results show that multiple annotators per panel are not necessarily needed.

References

- David Bau, Bolei Zhou, Aditya Khosla, Aude Oliva, and Antonio Torralba. 2017. [Network dissection: Quantifying interpretability of deep visual representations](#). *Preprint*, arXiv:1704.05796.
- Tolga Bolukbasi, Adam Pearce, Ann Yuan, Andy Coenen, Emily Reif, Fernanda Viégas, and Martin Wattenberg. 2021. [An interpretability illusion for bert](#). *Preprint*, arXiv:2104.07143.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. [Bert: Pre-training of deep bidirectional transformers for language understanding](#). *Preprint*, arXiv:1810.04805.
- Lukas Galke, Ansgar Scherp, Andor Diera, Fabian Karl, Bao Xin Lin, Bhakti Khera, Tim Meuser, and Tushar Singhal. 2022. [Are we really making much progress in text classification? a comparative review](#).
- Evan Hernandez, Sarah Schwettmann, David Bau, Teona Bagashvili, Antonio Torralba, and Jacob Andreas. 2022. [Natural language descriptions of deep visual features](#).
- Matthew Honnibal, Ines Montani, Sofie Van Landeghem, and Adriane Boyd. 2020. [spaCy: Industrial-strength natural language processing in python](#).
- Moinul Hossain, Sourav Rabi Das, Zikrul Shariar Ayon, Sadia Afrin Promi, Ahnaf Atef Choudhury, Shakila Rahman, and Jia Uddin. 2026. [Towards intelligent legal document analysis: Cnn-driven classification of case law texts](#).
- Alexander Liu and Samuel Yang. 2022. [Masked autoencoders as the unified learners for pre-trained sentence representation](#).
- Leland McInnes, John Healy, and James Melville. 2018. [Umap: Uniform manifold approximation and projection for dimension reduction](#).
- Sergio E. Zanotto and Segun Aroyehun. 2025. [Linguistic and embedding-based profiling of texts generated by humans and large language models](#).

A Results from the original paper

Datasets	Monotonic	Increasing	Decreasing
1	27.8%	13.9%	13.9%
2	22.7%	11.3%	11.4%
3	29.6%	14.7%	15.0%
4	27.0%	13.4%	13.6%
1, 2	7.7%	3.2%	3.2%
1, 3	9.8%	4.0%	4.1%
1, 4	9.2%	3.9%	3.9%
2, 3	10.9%	5.3%	5.4%
2, 4	7.4%	3.0%	2.9%
4, 3	9.9%	4.3%	4.3%
1, 2, 3	4.2%	1.8%	1.9%
1, 2, 4	3.0%	1.2%	1.2%
1, 3, 4	3.9%	1.7%	1.6%
2, 3, 4	4.2%	1.8%	1.8%
1, 2, 3, 4	1.9%	0.8%	0.8%

Table 6: Monotonicity data as presented in [Bolukbasi et al. \(2021\)](#), Table 3. Our results are numerically close to those obtained by the original authors. Dataset numbering corresponds to our [Table 3](#).